Daniel Guimarans

Principal AI Scientist | Heuristics & AI-Augmented Optimization for Supply Chain, Routing and Logistics

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Languages: English / Spanish / Catalan

SUMMARY

Principal AI Scientist with 20+ years in industry and academia, working on AI-augmented optimization for large-scale supply chain, routing and logistics problems. Technical lead for heuristics development at Lyric. Previously Principal Research Scientist (L7) at Amazon. Delivered optimization systems for transportation networks, port operations and airline crew planning, with cost savings exceeding $300MM. Methods span heuristics, mathematical programming, constraint programming, decomposition and simulation. PhD in Computer Science, 40+ peer-reviewed publications and 3 completed PhD supervisions.

EXPERIENCE

Feb 2026 – Present

Lyric Barcelona, Spain (remote)

Principal AI Scientist

- Technical lead for heuristics development across supply chain planning and routing products, guiding the teams in both streams.
- **Supply planning:** defined and delivered the heuristics-based solver (Python, Polars) used for large-scale problems on the Lyric platform, designed for solve speed and plan explainability.
- **Routing:** architected and built a heuristic solver (Rust) covering multiple problem variants. It approaches state-of-the-art solution quality in a fraction of the run time and scales to instances with tens of thousands of customers and multiple side constraints.
- **GPU optimization:** working with NVIDIA on evaluating the GPU-based cuOpt solver on large-scale routing problems.
- **Agentic optimization:** developing a framework that takes customer-specific optimization problems from description to a validated, deployable software package. Subject of a U.S. provisional patent application (2026).
- **Harness engineering:** autonomous research loops with controls that prevent agentic drift.

Feb 2020 – Jan 2026

Amazon Barcelona, Spain / Luxembourg

Apr 2022 – Jan 2026 Principal Research Scientist (L7), Barcelona

Feb 2020 – Apr 2022 Senior Research Scientist (L6), Luxembourg

- Optimization initiatives with cost savings exceeding $100MM annually.
- Principal scientist in a team that grew from under 10 to about 150 people. Sole Principal Scientist for most of that period, setting long-term research direction and science standards.
- **Network milk runs:** solved a tactical network design problem that had been open for years, cutting transportation cost while preserving or improving delivery speed through cut-off time (CPT) design. Directed model development from planner feedback gathered in multi-week workshops, working with the program management team to drive adoption against strong operator resistance to change.
- **Shipping with Amazon expansion:** science lead for integrating the business into middle-mile and last-mile operations, across 5+ VP organizations. Covered jurisdiction definition, consolidation center network design and volume scheduling to smooth production and prevent congestion. Combined pick-up and delivery routing with Amazon XL and Amazon Logistics. SVP-sponsored.
- **Workforce planning:** long-term labor planning across EU countries, maximizing stable permanent contracts under planning uncertainty. SVP-sponsored.
- **Other systems:** reverse logistics scheduler deployed in Europe, India and Japan; inbound network design and inventory allocation for the same-day delivery expansion. Advisor on long-term network planning, network design under uncertainty and middle-mile scheduling.
- **Science leadership:** mentored scientists across organizations on promotions and role transitions (e.g. SDE to Applied Scientist). Science bar evaluator for promotions to L6 and cross-org promotions to L7. 50+ hiring interviews. Organized the Amazon stream at EURO 2024 (Copenhagen), supporting Amazon Scholar recruitment.

Feb 2019 – Jan 2020

Monash University Melbourne, Australia

Research Fellow, Optimization and AI

- **Peak-hour demand shifting:** congestion-reduction system for Melbourne's metro network (500K+ daily commuters) with Public Transport Victoria. Trip patterns inferred from tap-on data.
- **Digital twins:** built an open-source discrete-event simulation engine based on augmented Petri nets for Queensland's coal rail network and ground handling at European airports (github.com/dguimarans/petri_net_simulator).
- Hybrid simulation-optimization methods for stochastic problems in aviation, logistics and transportation.

Oct 2015 – Jan 2019

Amsterdam University of Applied Sciences Amsterdam, Netherlands

Senior Researcher / Senior Lecturer

- Led research in optimization and simulation for aviation operations and disruption management.
- **Green taxi fleet:** data analysis and simulation-optimization of charging ranks for Amsterdam's green taxi fleet with Amsterdam City Council.
- **Program leadership:** developed and launched the Aviation Logistics curriculum and the Aviation Operations master program, coordinating a multidisciplinary team of researchers and lecturers.
- Supervised student projects hosted by KLM and Schiphol Airport. Taught 200+ students annually.

Mar 2013 – Oct 2015 **NICTA (now CSIRO's Data61)** Sydney, Australia
Researcher

- **Port Botany rail capacity:** simulation and scheduling study with NSW Ports. Showed that an AUD 210MM+ government-approved duplication of a section of the Port Botany freight rail line was unnecessary. NICTA Best Industry Project Award. Covered by The Sydney Morning Herald, The Australian and Lloyd's List.
- **Virgin Australia:** crew productivity and rostering optimization for the E190 fleet.
- **Canberra light rail:** traffic impact simulation with the ACT Government.
- Routing optimization systems for commercial fleet clients.

Apr 2005 – Feb 2013 **Universitat Autònoma de Barcelona** Barcelona, Spain
Sep 2012 – Feb 2013 Postdoctoral Researcher
Sep 2005 – Aug 2012 PhD Researcher and Adjunct Lecturer
Apr 2005 – Sep 2005 Research Assistant

- **PhD research:** hybrid algorithms for vehicle routing combining constraint programming, Lagrangian relaxation and metaheuristics, evaluated on academic benchmarks.
- **Applied research:** hybrid optimization and simulation for emergency services coordination, airline disruption management and airport ground handling.
- **ITSLogisim:** developed the distributed simulation platform behind the Simulation Suite during MSc research (2005–2007); the PhD algorithms became the basis of the Optimization Suite. Both suites were licensed to Digital Aeronautics Engineering Services (2010), the department's spin-off. Advised on commercial applications, including secure cash-collection routing and municipal waste collection.

PATENTS

- *Automated System for Producing and Validating Mathematical Optimization Software Packages.* U.S. Provisional Patent Application No. 64/103,812, filed Aug 2026 by ChainBrain Inc. (dba Lyric). Co-inventor.

EDUCATION

2012 **Ph.D. in Computer Science (Cum Laude)**
Universitat Autònoma de Barcelona
Thesis: Hybrid Algorithms for Solving Routing Problems

2007 **M.Sc. in Industrial Computer Science and Advanced Production Techniques**
Universitat Autònoma de Barcelona
Thesis: A Decision Support System for the Emergency Services Coordination Problem in a Road Accident. Sant Jordi Award for the best Catalan Master Thesis in Computer Science.

2005 **B.Sc. in Physics**
Universitat Autònoma de Barcelona
Minor in Applied Mathematics (2004)

SELECTED PUBLICATIONS AND TALKS

1,600+ citations, h-index 19 (Google Scholar). 30+ conference talks and invited seminars, including 2 keynotes. Full list of 40+ peer-reviewed publications at dguimarans.net.

Publications

- *Hierarchical Forecast Reconciliation on Networks: A Network Flow Optimization Formulation.* arXiv preprint, 2025.
- *Constraint-based Robust Planning and Scheduling of Airport Apron Operations through Simheuristics.* Annals of Operations Research, 2023.
- *A Stochastic Approach for Planning Airport Ground Support Resources.* International Transactions in Operational Research, 2022.
- *A Biased-Randomised Large Neighbourhood Search for the Two-Dimensional Vehicle Routing Problem with Backhauls.* European Journal of Operational Research, 2016.
- *A Bi-objective Approach for Scheduling Ground Handling Vehicles in Airports.* Computers & Operations Research, 2016.
- *Rich Vehicle Routing Problem: Survey.* ACM Computing Surveys, 2014.

Talks

- *Middle Mile Logistics at Scale: A Pragmatic Science Perspective.* Keynote on Amazon middle-mile science, OPTIMA-Con 2024, Melbourne.
- *A Scalable Solution to Joint Routing and Sorting Problems.* EURO 2024, Copenhagen (Amazon track).
- *Long-term Capacity and Topology Planning for a Large Delivery Network.* INFORMS Annual Meeting 2023, Phoenix (Amazon track).